Quiz 2 Practice Problem Set

MA4.101 Real Analysis, Monsoon 2026

Part A: Real numbers, order and suprema

Problem 1.

Let $a_n = \frac{1}{2} + \frac{(-1)^n}{n}$ for $n \geq 1$, so $a_1 = -\frac{1}{2}$, $a_2 = 1$, $a_3 = \frac{1}{6}, \ldots$

(a) Show (a_n) is a Cauchy sequence of rationals, and show $\mathrm{LIM}_{n\to\infty}\, a_n = \frac{1}{2}$.

(b) The sequence (a_n) is *not* positively bounded away from zero (since $a_1 < 0$). Construct a sequence (b_n) that is positively bounded away from zero and equivalent to (a_n), and using it conclude that $\mathrm{LIM}\, a_n$ is positive.

(c) State the general principle you used in (b).

(d) Let (c_n) be a Cauchy sequence of rationals with $c_n > 0$ for every n and $\mathrm{LIM}\, c_n \neq 0$. Prove that $\mathrm{LIM}\, c_n$ is positive.

Problem 2.

Let $x = \mathrm{LIM}\, a_n$ where $a_n = \dfrac{n+1}{2n}$.

(a) Show (a_n) is Cauchy and bounded away from zero (find a rational $c > 0$ with $|a_n| \geq c$ for all n).

(b) Show that $c_n = 1/a_n$ is Cauchy and find $\mathrm{LIM}\, c_n$ by proving (c_n) is equivalent to a constant sequence.

(c) Verify directly from the definition of multiplication that $x \cdot \mathrm{LIM}\, c_n = 1$.

Problem 3.

(a) Let (a_n) be a Cauchy sequence of rationals with $a_n \geq -\frac{1}{n}$ for all n. Prove $\mathrm{LIM}\, a_n \geq 0$.

(b) Let $(a_n), (b_n)$ be Cauchy sequences of rationals with $|a_n - b_n| \leq \frac{1}{n}$ for all n. Using part (a) only (not the definition of equivalence), prove $\mathrm{LIM}\, a_n = \mathrm{LIM}\, b_n$.

(c) Let $s_n = \sum_{k=1}^{n} \dfrac{1}{k(k+1)}$. Show $s_n = 1 - \frac{1}{n+1}$ and compute $\mathrm{LIM}\, s_n$.

(d) Let x be a real with $x \geq 0$ and $x \leq \frac{1}{n}$ for every $n \geq 1$. Prove $x = 0$. Deduce: if $x \leq y + \frac{1}{n}$ for every $n \geq 1$ then $x \leq y$.

Problem 4.

For a real x let $\lfloor x \rfloor$ denote the unique integer m with $m \leq x < m + 1$. Prove:

(a) $\lfloor x + k \rfloor = \lfloor x \rfloor + k$ for every integer k.

(b) $\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor \leq \lfloor x \rfloor + \lfloor y \rfloor + 1$.

(c) If x is not an integer then $\lfloor -x \rfloor = -\lfloor x \rfloor - 1$.

(d) For every $n \geq 1$, $0 \leq x - \dfrac{\lfloor nx \rfloor}{n} < \dfrac{1}{n}$.

(e) (Existence of the integer part.) Prove by induction on $n \in \{0, 1, 2, \dots\}$ the statement $P(n)$: "for every real x with $0 \leq x < n$ there is an integer m with $m \leq x < m + 1$". Conclude that every real $x \geq 0$ has an integer part.

Problem 5.

Let x be a real number and let $S_x = \{q \in \mathbb{Q} : q < x\}$. You may use that between any two reals there is a rational.

(a) Prove S_x is nonempty and bounded above, and that $\sup S_x = x$.

(b) Prove S_x has no largest element, even when x is rational.

(c) Prove: for reals x, y, $S_x \subseteq S_y \iff x \leq y$.

(d) Deduce: $S_x = S_y \iff x = y$.

Problem 6.

Let $E = \{q \in \mathbb{Q} : q^2 < 5\}$.

(a) Prove that no rational number q satisfies $q^2 = 5$.

(b) Show that E is nonempty and that every positive rational r with $r^2 > 5$ is an upper bound of E. Show also that every upper bound of E in $\mathbb{Q}$ is ≥ 2.

(c) For a positive rational r put $s = \dfrac{3r + 5}{r + 3}$. Show that s is rational and positive, $s^2 - 5 = \dfrac{4(r^2 - 5)}{(r + 3)^2}$, and $s - r = \dfrac{5 - r^2}{r + 3}$.

(d) Let r be any rational upper bound of E. Show $r^2 > 5$, and construct a smaller rational upper bound. Conclude that E has no least upper bound in $\mathbb{Q}$.

Problem 7.

For a nonempty set $A \subseteq \mathbb{R}$ that is bounded above, let $U(A) = \{u \in \mathbb{R} : a \leq u \text{ for all } a \in A\}$ be the set of its upper bounds.

(a) Prove $U(A) = [\sup A, \infty)$.

(b) Prove: $s = \sup A$ if and only if (i) $s \in U(A)$ and (ii) for every $\varepsilon > 0$ there is $a \in A$ with $a > s - \varepsilon$.

(c) For nonempty bounded-above sets A, B prove $U(A \cup B) = U(A) \cap U(B)$, and deduce $\sup(A \cup B) = \max\{\sup A, \sup B\}$.

(d) Prove $\{x \in \mathbb{R} : x \leq \frac{1}{n} \text{ for every } n \geq 1\} = (-\infty, 0]$.

Problem 8.

Let A be a nonempty set of positive reals and let $A^{-1} = \{1/a : a \in A\}$.

(a) If $m = \inf A > 0$, prove A^{-1} is bounded above and $\sup A^{-1} = 1/m$.

(b) If $\inf A = 0$, prove A^{-1} is not bounded above, i.e. $\sup A^{-1} = +\infty$ in $\mathbb{R}^*$.

(c) Use (a) and (b) to find $\sup\left\{\dfrac{n}{n+1} : n \geq 1\right\}$ and $\sup\left\{n + \dfrac{1}{n} : n \geq 1\right\}$.

Problem 9.

Let $S = \{x \in \mathbb{R} : x \geq 0,\ x^3 < 2\}$.

(a) Show S is nonempty and bounded above by 2, so $s = \sup S$ exists with $1 \leq s \leq 2$.

(b) Suppose $s^3 < 2$. Let $\delta = \min\{1, \frac{2-s^3}{20}\}$ and show $(s+\delta)^3 < 2$. Derive a contradiction.

(c) Suppose $s^3 > 2$. Let $\delta = \dfrac{s^3-2}{6s^2}$. Show $0 < \delta < s$ and $(s-\delta)^3 > 2$, and deduce that $s - \delta$ is an upper bound of S. Derive a contradiction.

(d) Conclude $s^3 = 2$, and show that s is the only positive real with cube equal to 2.

Hint for (c): for $0 \leq \delta \leq s$, $(s-\delta)^3 = s^3 - 3s^2\delta + \delta^2(3s-\delta) \geq s^3 - 3s^2\delta$.

Part B: Sequences, limits and monotone sequences

Problem 10.

(a) Prove from the definition (find N in terms of ε) that $\dfrac{5n-3}{2n+7} \to \dfrac{5}{2}$.

(b) Prove from the definition that $\sqrt{n^2+1} - n \to 0$.

(c) Let $a_n \geq 0$ for all n and $a_n \to L$. Prove $L \geq 0$.

(d) Let $a_n \to L$ and $b_n \to L'$ with $a_n \leq b_n$ for all n. Prove $L \leq L'$. (*Hint:* suppose $L > L'$ and take $\varepsilon = \frac{L-L'}{3}$.)

(e) True or false: if $x_n < y_n$ for all n and both sequences converge, then $\lim x_n < \lim y_n$. Justify.

Problem 11.

Let $b_n = \dfrac{a_n + a_{n+1}}{2}$.

(a) If $a_n \to L$, prove $b_n \to L$.

(b) Show the converse is false by giving a divergent (a_n) for which (b_n) converges.

(c) If (a_n) is *increasing* and (b_n) converges to L, prove (a_n) converges to L.

Problem 12.

(a) Let $a_n = \sum_{k=1}^{n} \dfrac{1}{\sqrt{k}}$. Prove $|a_{2n} - a_n| \geq \frac{1}{2}$ for every n, and conclude that (a_n) is not Cauchy and hence does not converge.

(b) Let $b_n = \sum_{k=1}^{n} \dfrac{1}{k^2}$. Prove that for $m > n \geq 1$, $|b_m - b_n| < \frac{1}{n}$, and conclude that (b_n) converges.

(c) Show that $1 \leq \lim b_n \leq 2$.

(d) True or false: if (a_n) is a Cauchy sequence of rationals and $x = \mathrm{LIM}\, a_n$, then (a_n) converges to x as a sequence of reals.

Hint for (b): for $k \geq 2$, $\frac{1}{k^2} \leq \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}$.

Problem 13.

(a) A student argues: "$n \cdot \frac{1}{n}$: $\lim n$ does not exist, so $\lim n \cdot \frac{1}{n}$ does not exist." Explain precisely what is wrong, and give the correct limit.

(b) Compute $\lim_{n\to\infty} \left(\sqrt{4n^2+n} - 2n\right)$.

(c) Compute $\lim_{n\to\infty} \dfrac{1+2+\cdots+n}{n^2}$.

(d) Compute $\lim_{n\to\infty} \dfrac{(n+1)(2n+3)}{n^2+5}$.

(e) Using only the ε-N definition, prove that if $a_n \to a$ then $a_n^3 \to a^3$. (You will need that (a_n) is bounded: prove that as well.)

(f) True or false: if $a_n^2 \to 4$, then $a_n \to 2$ or $a_n \to -2$.

Problem 14.

In $\mathbb{R}^* = \mathbb{R} \cup \{-\infty, +\infty\}$ (ordered by $-\infty < x < +\infty$ for real x) find sup and inf of each set, with brief justification.

(i) $E_1 = \{(-1)^n n : n \geq 1\}$.

(ii) $E_2 = \{\frac{1}{n} : n \geq 1\} \cup \{-\infty\}$.

(iii) $E_3 = \{x \in \mathbb{R} : x < 3\} \cup \{+\infty\}$.

(iv) $E_4 = \varnothing$.

(v) $E_5 = \{x \in \mathbb{R}^* : x \geq 5\}$.

Then prove: *every* subset of $\mathbb{R}^*$ has a supremum in $\mathbb{R}^*$.

Problem 15.

Consider $a_{n+1} = \dfrac{1+a_n^2}{2}$.

(a) Take $a_1 = 0$. Show $0 \leq a_n < 1$ for all n and that (a_n) is increasing. Conclude it converges and find the limit.

(b) Take $a_1 = 2$. Show (a_n) is increasing, and prove that it diverges (so $\sup a_n = +\infty$).

(c) Let (a_n) be *any* increasing sequence and suppose the subsequence (a_{2n}) converges to L. Prove that the whole sequence (a_n) converges to L.

Hint: $a_{n+1} - a_n = \frac{1}{2}(a_n - 1)^2$.

Problem 16.

Let $a_n = \dfrac{n^2}{2^n}$.

(a) Show $\dfrac{a_{n+1}}{a_n} = \dfrac{(n+1)^2}{2n^2}$. Show this ratio exceeds 1 for $n = 1, 2$ and is at most 1 for all $n \geq 3$.

(b) Deduce $\sup_n a_n = \max_n a_n = \frac{9}{8}$.

(c) Prove $a_n \to 0$, and hence $\inf_n a_n = 0$ (not attained).

Part C: limsup, liminf, squeeze, subsequences, BW

Problem 17.

Recall $a_N^+ = \sup_{n\ge N} a_n$, $a_N^- = \inf_{n\ge N} a_n$, $\limsup a_n = \inf_N a_N^+$, $\liminf a_n = \sup_N a_N^-$. For each sequence find sup, inf, lim sup, lim inf (in $\mathbb{R}^*$) and the set of real limit points.

(i) $a_n = r_n + \frac{1}{n}$, where $r_n \in \{0, \frac{1}{3}, \frac{2}{3}\}$ is $\frac{1}{3}(n \bmod 3)$.

(ii) $b_n = (-1)^n \left(1 - \frac{1}{n}\right)$.

(iii) $c_n = n$ if $3 \mid n$, and $c_n = (-1)^n$ otherwise.

(iv) For the sequence (b_n) in (ii), prove *by two inclusions* that the set of limit points is exactly $\{-1, 1\}$.

Problem 18.

(a) Write, using quantifiers, the negation of "$a_n \to L$" and the negation of "L is a limit point of (a_n)".

(b) Prove: $a_n \not\to L$ if and only if there are $\varepsilon > 0$ and a subsequence (a_{n_j}) with $|a_{n_j} - L| > \varepsilon$ for all j.

(c) For a real L and $\varepsilon > 0$ let $I_\varepsilon(L) = \{n \ge 1 : |a_n - L| \le \varepsilon\}$. Prove: L is a limit point of (a_n) if and only if $I_\varepsilon(L)$ is infinite for every $\varepsilon > 0$.

(d) Use (c) to prove: if $a_n \to L$ then L is a limit point and it is the *only* limit point of (a_n).

Problem 19.

Let $(a_n), (b_n)$ be bounded real sequences.

(a) Prove $\liminf(-a_n) = -\limsup a_n$.

(b) If $a_n \le b_n$ for all n, prove $\limsup a_n \le \limsup b_n$ and $\liminf a_n \le \liminf b_n$.

(c) If $a_n \to L$, prove $\limsup(a_n + b_n) = L + \limsup b_n$.

(d) Give an example where $\limsup(a_n b_n) \ne \limsup a_n \cdot \limsup b_n$.

(e) Let c be a real number. Prove: (i) if $c > \limsup a_n$ then $a_n < c$ for all large n; (ii) if $c < \limsup a_n$ then for every N there is $n \ge N$ with $a_n > c$.

Fact you may use: for a bounded nonempty set A, $\inf(-A) = -\sup A$.

Problem 20.

(a) Show $\dfrac{n + \cos n}{n + 1} \to 1$.

(b) Let $y_n = \displaystyle\sum_{k=1}^{n} \frac{n}{n^2 + k}$. Show $y_n \to 1$.

(c) Suppose $|a_{n+1}| \le \frac{3}{4}|a_n|$ for all $n \ge 2$. Prove $a_n \to 0$ (you may use $(\frac{3}{4})^n \to 0$). Apply this to $a_n = \dfrac{n}{2^n}$.

(d) A student "proves" the squeeze test: "If $a_n \le b_n \le c_n$ and $\lim a_n = \lim c_n = \ell$, take limits to get $\ell \le \lim b_n \le \ell$." Identify the gap, and repair the proof in two different ways.

Problem 21.

Let (a_n) be a bounded real sequence.

(a) Prove: (a_n) converges if and only if for every $\varepsilon > 0$ there is N with $a_N^+ - a_N^- \le \varepsilon$.

(b) Suppose that for every $\varepsilon > 0$ there is N such that $a_n \le a_m + \varepsilon$ whenever $n \ge m \ge N$. Prove $\limsup a_n \le \liminf a_n + \varepsilon$ for every $\varepsilon > 0$, and deduce that (a_n) converges.

(c) Show that every bounded decreasing sequence satisfies the hypothesis of (b), and that $a_n = \frac{(-1)^n}{n}$ satisfies it too.

You may use: $\liminf a_n \le \limsup a_n$ for bounded sequences, and that a bounded sequence converges if $\limsup a_n = \liminf a_n$.

Problem 22.

(a) If $a_{2n} \to L$ and $a_{2n-1} \to L$, prove $a_n \to L$.

(b) Give a bounded sequence with $a_{2n} \to L$ for some L such that (a_n) diverges.

(c) Let (a_n) be bounded with $\limsup a_n = 3$ and $\liminf a_n = 1$. Prove: (i) there is a subsequence converging to 3, and one converging to 1; (ii) every convergent subsequence has its limit in $[1, 3]$.

(d) Must every point of $[1, 3]$ be a limit point of such a sequence? Justify.

(e) True or false: (i) if $a_n \to L$, then every subsequence of (a_n) converges to L; (ii) every bounded sequence of reals has at least one limit point; (iii) if (a_n) is bounded, then $\limsup a_n$ is a limit point of (a_n).

Problem 23.

Call an index n a *peak* of a real sequence (a_n) if $a_n \ge a_m$ for every $m > n$.

(a) If there are infinitely many peaks, show (a_n) has a decreasing subsequence.

(b) If there are only finitely many peaks, show (a_n) has a strictly increasing subsequence.

(c) Deduce that every real sequence has a monotone subsequence.

(d) Deduce the Bolzano-Weierstrass theorem: every bounded real sequence has a convergent subsequence.

(e) Is the conclusion of (d) still true for an unbounded sequence? Give an example and a counterexample.